

Chapter 7 Exercises

Q1

The figure shows a singly-linked list. “Tokyo” is at the first element of the list, and this pointer contains the address of the next element. In addition, “Nagoya” is at the last element of the list, and this pointer contains 0. Which of the following is the process that inserts “Shizuoka” between “Atami” and “Hamamatsu”? Here, “Shizuoka” is located at address 150.

Pointer to the first element	Address	Element	Pointer
10	10	Tokyo	50
	30	Nagoya	0
	50	Shin Yokohama	90
	70	Hamamatsu	30
	90	Atami	70
	150	Shizuoka	

- Setting the pointer of Shizuoka to be 50 and the pointer of Hamamatsu to be 150
- Setting the pointer of Shizuoka to be 70 and the pointer of Atami to be 150
- Setting the pointer of Shizuoka to be 90 and the pointer of Hamamatsu to be 150
- Setting the pointer of Shizuoka to be 150 and the pointer of Atami to be 90

Q2

Which of the following is an appropriate description of a stack?

- Data that is stored last can be extracted first.
- Data that is stored first can be extracted first.
- By tracing from the first element, data can be extracted in sequence.
- By converting a search key into an address, data can be extracted.

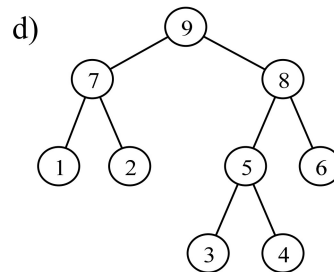
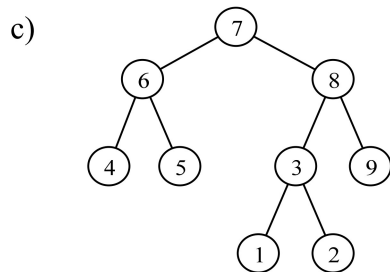
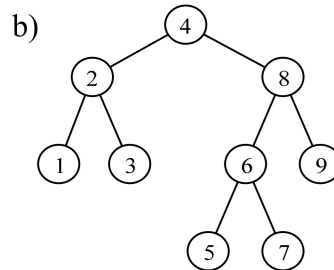
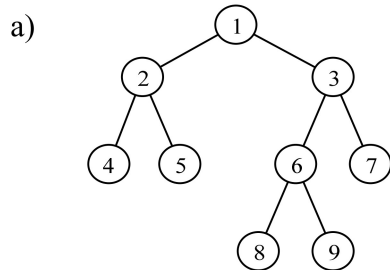
Q3

In the reverse Polish notation, expression “ $X = (A - B) \times C$ ” is represented as “ $XAB - C \times =$ ”. Which of the following is the representation of expression “ $X = (A + B) \times (C - D \div E)$ ” in the reverse Polish notation?

- | | |
|--------------------------------|---------------------------------|
| a) $XAB + CDE \div - \times =$ | b) $XAB + C - DE \div \times =$ |
| c) $XAB + EDC \div - \times =$ | d) $XBA + CD - E \div \times =$ |

Q4

Which of the following is an appropriate binary search tree? Here, numbers 1 through 9 show the value of each node.

**Q5**

Which of the following is the programming technique that is based on the principle of using only Sequence, Selection, and Iteration as the basic structure for developing an easy-to-understand program?

- a) Functional programming
- b) Structured programming
- c) Parallel programming
- d) Logic programming

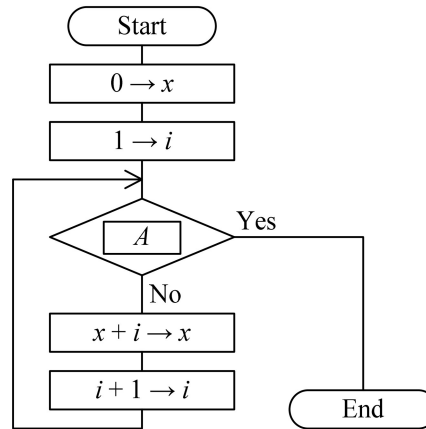
Q6

Which of the following is an appropriate explanation of the basic structure of an algorithm?

- a) “REPEAT UNTIL iteration” is used for representing an iteration process with a nested structure.
- b) “IF THEN ELSE selection” executes two processes in parallel.
- c) “CASE selection” is used to return to the process just prior to the selection.
- d) “DO WHILE iteration” may not execute the iteration process even once.

Q7

The following flowchart shows an algorithm that determines the sum (i.e. “ $1+2+\dots+N$ ”) of integer numbers from 1 through N ($N \geq 1$) and inserts the results into variable x . Which of the following is an expression that fits into blank A in the flowchart?



- a) $i = N$ b) $i < N$ c) $i > N$ d) $x > N$

Q8

Which of the following is an appropriate description of binary search?

- a) Data is correctly searched only when it is sorted in the ascending order.
 b) Data is correctly searched only when it is sorted in the ascending or descending order.
 c) Data is efficiently searched when data is sorted in the ascending or descending order.
 d) Data is correctly searched only when the number of data items is an even number.

Q9

Which of the following is the correct combination of a search method and the order of its execution time? Here, the number of data items to be searched is n , and probability of collision of hash values (i.e., values becoming the same) is so small that it can be ignored.

	Binary search	Linear search	Hash search
a)	$\log_2 n$	n	1
b)	$n \log_2 n$	n	$\log_2 n$
c)	$n \log_2 n$	n^2	1
d)	n^2	1	n

Q10

When n data items are sorted with selection sort, how many times are data items compared?

- | | |
|----------------------------|----------------------------|
| a) $n \times \log n$ | b) $n \times (n+1) \div 4$ |
| c) $n \times (n-1) \div 2$ | d) n^2 |

Q11

Which of the following is an appropriate explanation of heap sort?

- It sorts the substring where elements are extracted at regular intervals, and it furthermore extracts the substring where the interval is reduced and sorts it.
- After an intermediate reference value is decided, it divides elements in a segment where values that are larger than this value are assembled and in a segment where values that are smaller are assembled. Then, a similar process is repeated in the respective segment.
- After adjacent elements are compared, if the order of magnitude is reverse, it exchanges these elements. Then, this process is repeated.
- The unsorted part is configured as an ordered tree, and the maximum value or the minimum value is extracted from there and moved to the sorted part. This operation is repeated, and the unsorted part is gradually reduced.

Q12

Which of the following is an appropriate explanation of the Boyer-Moore method?

- a) It is an aggregation algorithm that reads the records that are sorted with key items in order, categorize the records that have the same value of key items as a same group, and calculates the sum of the records.
- b) It is an n -th order solution algorithm that is used when an approximate solution is found, approximates a true solution by calculating a tangential equation in an approximate solution.
- c) It is a shortest path search algorithm that decides the minimum cost up to all nodes by finalizing the node that has the minimum cost among the nodes whose paths are clarified.
- d) It is a string pattern matching algorithm that decides the amount of movement of a partial string on the basis of whether the last character of the partial string that is extracted from the search target is present in the string to be searched for or not.